

DeepSafe: Hyperbolic Optimal Transport Projection for Intent-Aware LLM Safety Classification

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Abstract

Current safety classifiers for large language models (LLMs) predominantly rely on surface-level keyword matching and semantic similarity, conflating malicious intent (e.g., “How to build a bomb to hurt people”) with benign educational discussion (e.g., “What are the dangers of building bombs?”). This *intent blindness* leads to over-censorship of legitimate content and erodes user trust. We propose **DeepSafe**, a geometrically principled projection framework that integrates three key innovations: (i) **hyperbolic neural projection** onto the Poincaré ball to naturally encode the hierarchical safety taxonomy (safe $\rightarrow$ pseudo-harmful $\rightarrow$ malicious), (ii) **optimal transport regularization** via Sinkhorn divergence to achieve distribution-level class separation beyond instance-level contrastive loss, and (iii) **learnable prototype anchors** with hierarchical margins that explicitly structure the representation space around class centroids. DeepSafe further employs spectral decorrelation to eliminate feature redundancy and counterfactual pair repulsion to target intent-blindness. With only 0.87M–4.3M trainable parameters, DeepSafe operates as a drop-in projection head over frozen LLM embeddings. On a 257K-sample safety classification benchmark, DeepSafe achieves up to 86.9% accuracy—competitive with state-of-the-art guard models including Qwen3Guard-Gen-4B, Granite Guardian 3.x, and ShieldGemma 2. Our analysis demonstrates that hyperbolic geometry alone accounts for over 50% improvement in hierarchical class separation over Euclidean projections, while the combined framework achieves generalization that rivals models 10–100 $\times$ its size. The learned representations enable interpretable per-intent safety analysis and substantially reduce false positives on legitimate educational content.

Paper Checklist

1. **Claims:** DeepSafe improves safety classification through hyperbolic geometry, optimal transport, and prototype learning. Claims are supported by experiments in Section 5 with quantitative metrics and comparisons against SOTA guard models.
2. **Limitations:** Discussed in Section 6. Hyperbolic curvature is treated as a fixed hyperparameter rather than learned. OT regularization incurs $\mathcal{O}(B^2)$ complexity for batch size B .
3. **Theory & Assumptions:** The core assumption is that safety labels form a hierarchy best captured by hyperbolic geometry. This is formalized in Section 4.1 and validated empirically.
4. **Open Source:** Code, trained weights, and evaluation scripts will be released upon acceptance.
5. **Data:** We use a 257K-sample dataset with keyword-based intent labels. Details in Section 5.1.
6. **Experimental Results:** All results are reproducible with fixed seeds (42). Training logs, evaluation reports, and raw metrics included.

1 Introduction

Content safety has become a critical concern as large language models (LLMs) are deployed at scale in user-facing applications. Current safety classifiers typically operate as binary filters on model inputs and outputs, using keyword blocklists, embedding similarity thresholds, or fine-tuned classifiers to flag potentially unsafe content [4, 8]. While effective at catching overtly harmful content, these approaches suffer from a fundamental limitation: they are **intent-blind**.

Consider two queries sharing the same topic keywords: (1) “How do I build a bomb to hurt people?” and (2) “What are the dangers of building bombs?” A conventional safety classifier, trained to associate the token sequence ‘build’ + ‘bomb’ with unsafe content, will flag both queries as harmful. This conflates malicious intent with benign educational or safety-related discussion, leading to three concrete harms: (i) **over-censorship** of legitimate user queries, (ii) **erosion of user trust** when the system appears overly restrictive, and (iii) **missed opportunities** for safety interventions that could redirect users toward helpful resources.

We argue that the core problem is *representation collapse*: existing embedding spaces (e.g., Qwen3-Embedding [11]) and the classifiers built atop them lack the structural priors needed to distinguish malicious intent from benign discussion of the same topic. As shown in Figure 1 (left), the original embedding space

exhibits near-zero silhouette score (<0.01), with safe, pseudo-harmful, and malicious content completely overlapping.

To address this, we propose **DeepSafe**, a geometrically principled projection framework that learns to reorganize frozen LLM embeddings into an intent-aware representation space through three synergistic innovations:

1. **Hyperbolic Neural Projection:** Safety labels form a natural hierarchy (Safe $\rightarrow$ Pseudo-Harmful $\rightarrow$ Malicious) where each level subsumes the semantics of the previous. Euclidean geometry poorly captures such tree-like structure, as volume grows polynomially with radius while the number of hierarchical descendants grows exponentially. We project embeddings onto the Poincaré ball model of hyperbolic geometry [15], where negative curvature naturally encodes hierarchical organization. Möbius matrix-vector multiplication and the logarithmic/exponential maps enable neural network operations directly within the hyperbolic manifold.
2. **Optimal Transport Regularization:** Beyond instance-level contrastive learning, we introduce Sinkhorn divergence [17] as a distribution-level objective that directly optimizes the distance between empirical class distributions, ensuring global geometric separation rather than just local pairwise relationships.
3. **Learnable Prototype Anchors:** Explicit class centroids with hierarchical margin constraints (Safe $\leftrightarrow$ Malicious: 0.5, Safe $\leftrightarrow$ Pseudo-Harmful: 0.25) structure the representation space around interpretable safety regions, preventing the representation drift that can occur with purely contrastive objectives.

DeepSafe further employs spectral decorrelation [16] to eliminate feature redundancy, and counterfactual pair repulsion to directly target intent-blindness by maximizing the distance between topic-matched utterances with opposing intents.

DeepSafe is remarkably parameter-efficient: the projection head contains only 0.87M parameters (0.6B backbone) or 4.3M parameters (8B backbone), operates on frozen embeddings (1024-dim or 4096-dim), and outputs 256-dimensional intent-aware representations. This makes it practical to deploy as a drop-in preprocessing step before any downstream safety classifier.

Our contributions are:

- We formalize the intent-blindness problem in LLM safety classification and propose a hierarchical intent taxonomy as a structural prior best captured by hyperbolic geometry.
- We introduce DeepSafe, a geometrically principled framework combining hyperbolic projection, optimal transport, prototype learning, and spectral decorrelation for safety representation learning.
- Through extensive experiments on 257K samples with two embedding backbones (0.6B and 8B), we demonstrate that DeepSafe (i) achieves up to 86.92% accuracy matching or surpassing guard models 10–100 $\times$ its size, (ii) improves representation quality by over 50 $\times$ in silhouette score and 10 $\times$ in separation ratio, and (iii) enables interpretable per-intent accuracy analysis with 79.6% accuracy on the hardest pseudo-harmful category.

2 Related Work

2.1 LLM Safety Classification

Safety classification for LLMs has been approached from multiple angles. Pre-deployment safety alignment methods such as RLHF [10], DPO [12], and constitutional AI [1] aim to make models generate safer outputs but do not provide explicit safety judgments. Post-hoc safety filters use separate classifier models to detect harmful content in inputs and outputs [4, 8]. These classifiers are typically trained as binary discriminators on embedding representations, achieving strong overall accuracy but lacking the ability to distinguish malicious intent from benign discussion. Our work complements these approaches by learning an intent-aware representation space that can be used as input to any downstream classifier.

2.2 Contrastive Representation Learning

Contrastive learning has emerged as a powerful paradigm for learning structured representations. SimCLR [2] and its variants learn representations by maximizing agreement between augmented views of the same sample. Supervised contrastive learning [7] extends this to labeled data by pulling together samples of the same

48 class. In NLP, contrastive learning has been applied to sentence embeddings [3] and retrieval [5]. Our hierarchical contrastive loss extends supervised contrastive learning to operate on multiple granularities of labels simultaneously (binary safety and fine-grained intent), which to our knowledge is a novel application in the safety domain.

2.3 Intent Recognition in NLP

Intent recognition has been extensively studied in task-oriented dialogue systems [13] and more recently for detecting harmful intent in online content [9]. However, these approaches typically treat intent as a standalone classification task rather than integrating it into the representation learning process. DeepSafe differs by using intent labels as a structural prior during representation learning, such that the embedding space itself encodes intent distinctions.

2.4 Counterfactual and Causal Approaches

Counterfactual data augmentation has been used to improve model robustness [6]. In safety contexts, counterfactual pairs—utterances that differ only in intent while preserving topic—have been proposed as evaluation benchmarks [14]. We extend this idea by incorporating counterfactual pairs directly into the training objective through our pair repulsion loss.

3 Method

3.1 Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i^b, y_i^I)\}_{i=1}^N$ be a dataset of N utterances, where $\mathbf{x}_i$ is the text, $y_i^b \in \{0, 1\}$ is a binary safety label (0 = safe, 1 = unsafe), and $y_i^I \in \{0, 1, 2\}$ is a fine-grained intent label following our hierarchical taxonomy:

$$y^I = \begin{cases} 0 & \text{Safe: unambiguously benign content} \\ 1 & \text{Pseudo-Harmful: discussion of sensitive topics with legitimate purpose} \\ 2 & \text{Malicious: explicitly harmful or malicious content} \end{cases} \quad (1)$$

The hierarchical relationship is: Safe ($y^I = 0$) maps to $y^b = 0$; Pseudo-Harmful ($y^I = 1$) maps to $y^b = 0$ (safe in intent); Malicious ($y^I = 2$) maps to $y^b = 1$ (unsafe).

Let $f_\theta(\mathbf{x}) = \mathbf{e} \in \mathbb{R}^{1024}$ be a frozen pre-trained embedding model (we use Qwen3-Embedding-0.6B [11]). Our goal is to learn a projection head $g_\phi : \mathbb{R}^{1024} \rightarrow \mathbb{S}^{255}$ that maps embeddings onto the unit hypersphere in $\mathbb{R}^{256}$, such that the projected representations $\mathbf{z}_i = g_\phi(f_\theta(\mathbf{x}_i))$ satisfy: (a) binary safety classes are linearly separable, (b) intent classes form distinct clusters, and (c) counterfactual pairs with shared keywords but different intents are maximally separated.

3.2 Projection Head Architecture

DeepSafe employs a geometrically principled projection head $g_\phi : \mathbb{R}^D \rightarrow \mathbb{S}^{255}$ that combines three complementary mechanisms:

Hyperbolic Neural Projection. Safety labels form a natural hierarchy (Safe $\rightarrow$ Pseudo-Harmful $\rightarrow$ Malicious), where each level subsumes the semantics of the previous. Euclidean geometry poorly captures such hierarchical structure, as the volume of Euclidean space grows polynomially with radius, while the number of hierarchical descendants grows exponentially [15]. We therefore project embeddings into the Poincaré ball model of hyperbolic geometry, where the negative curvature $c = 1.0$ causes volume to grow exponentially, naturally encoding tree-like hierarchies.

Given a frozen embedding $\mathbf{e} \in \mathbb{R}^D$, we first project to a tangent space $\mathbf{t} = \text{LayerNorm}(\mathbf{W}_{\text{in}}\tilde{\mathbf{e}})$, where $\tilde{\mathbf{e}} = \mathbf{e}/\|\mathbf{e}\|_2$. The exponential map $\exp_{\mathbf{0}}^c(\mathbf{t})$ lifts $\mathbf{t}$ onto the Poincaré ball $\mathbb{B}_c^d$:

$$\exp_{\mathbf{0}}^c(\mathbf{t}) = \tanh(\sqrt{c}\|\mathbf{t}\|) \frac{\mathbf{t}}{\sqrt{c}\|\mathbf{t}\|} \quad (2)$$

Within the hyperbolic space, we apply Möbius matrix-vector multiplication $\mathbf{W} \otimes_c \mathbf{x} = \exp_0^c(\mathbf{W} \log_0^c(\mathbf{x}))$, followed by Möbius addition of a bias. The logarithmic map $\log_0^c$ then brings features back to Euclidean space for downstream processing:

$$\log_0^c(\mathbf{x}) = \frac{\text{arctanh}(\sqrt{c} \|\mathbf{x}\|)}{\sqrt{c} \|\mathbf{x}\|} \mathbf{x} \quad (3)$$

The hyperbolic pathway captures hierarchical safety structure, while the distance to origin $\|\mathbf{x}\|_{\mathbb{B}}$ encodes the “severity” of content in the safety taxonomy.

Residual Euclidean Pathway. To preserve fine-grained semantic information that may be lost in the hyperbolic projection, we maintain a parallel Euclidean pathway: $\mathbf{r} = \mathbf{W}_{\text{res}} \tilde{\mathbf{e}}$. The hyperbolic and Euclidean features are summed and processed through a lightweight MLP with GELU activation, dropout, and layer normalization, producing the final L2-normalized output $\mathbf{z} \in \mathbb{S}^{255}$.

Spectral Decorrelation. We apply a Barlow Twins-style [16] regularization that penalizes off-diagonal elements of the feature correlation matrix, promoting diverse, non-redundant feature dimensions:

$$\mathcal{L}_{\text{decorr}} = \lambda_d \sum_{i \neq j} \left(\frac{1}{B} \mathbf{Z}^\top \mathbf{Z} \right)_{ij}^2 \quad (4)$$

3.3 Optimal Transport Class Regularization

Standard contrastive loss operates at the instance level, which can be noisy and sensitive to outliers. We introduce a complementary distribution-level objective based on entropic optimal transport. For each class pair (a, b) , the Sinkhorn divergence [17] provides a principled distance between the empirical distributions of their projected representations:

$$\mathcal{L}_{\text{OT}} = \sum_c S_\varepsilon(\mathbf{Z}_c, \mathbf{Z}_c) - \lambda_{\text{inter}} \sum_{a < b} S_\varepsilon(\mathbf{Z}_a, \mathbf{Z}_b) \quad (5)$$

where S_ε is the Sinkhorn divergence computed via the Sinkhorn-Knopp algorithm with entropic regularization ε . The first term promotes intra-class compactness, while the second term encourages inter-class separation at the distribution level.

3.4 Hierarchical Contrastive Loss

We retain the hierarchical contrastive loss [7] operating at two granularities:

$$\mathcal{L}_{\text{hier}} = \mathcal{L}_{\text{bin}} + \alpha \cdot \mathcal{L}_{\text{intent}} \quad (6)$$

where $\mathcal{L}_{\text{bin}}$ contrasts safe vs. unsafe samples, and $\mathcal{L}_{\text{intent}}$ contrasts the three fine-grained intent classes (Safe, Pseudo-Harmful, Malicious).

3.5 Learnable Prototype Anchors

To provide explicit structural guidance, we maintain learnable prototype vectors $\{\mathbf{p}_k\}_{k=0}^2$ in the projected space, one per intent class. The prototype loss applies hierarchical margins reflecting the intent taxonomy:

$$\mathcal{L}_{\text{proto}} = \underbrace{-\frac{1}{B} \sum_i \cos(\mathbf{z}_i, \mathbf{p}_{y_i})}_{\text{attraction}} + \underbrace{\lambda_p \frac{1}{B} \sum_i \sum_{c \neq y_i} \max(0, \cos(\mathbf{z}_i, \mathbf{p}_c) - m_{y_i, c})}_{\text{hierarchical repulsion}} \quad (7)$$

where the margin matrix $m_{y,c}$ encodes the semantic distance between intent classes: Safe $\leftrightarrow$ Malicious uses the largest margin (0.5), Safe $\leftrightarrow$ Pseudo-Harmful uses a smaller margin (0.25), and Pseudo-Harmful $\leftrightarrow$ Malicious uses an intermediate margin (0.35).

Table 1: Classification performance comparison. All methods use Logistic Regression as the downstream classifier.

Method	Accuracy	F1 (Macro)	ROC AUC
<i>Baselines (Original Frozen Embeddings, 1024-dim)</i>			
Logistic Regression	0.8123	0.8116	0.8928
Linear SVM	0.8126	0.8120	0.8928
Gaussian Naive Bayes	0.7137	0.7128	0.7856
KNN ($k = 5$)	0.8096	0.8080	0.8818
<i>Our Methods (Projected Embeddings, 256-dim)</i>			
DeepSafe + LR (Ours)	0.8431	0.8423	0.9177

3.6 Full Training Objective

$$\mathcal{L}_{\text{DeepSafe}} = \mathcal{L}_{\text{hier}} + \lambda_{\text{ot}} \mathcal{L}_{\text{OT}} + \lambda_p \mathcal{L}_{\text{proto}} + \mathcal{L}_{\text{decorr}} \quad (8)$$

We use $\lambda_{\text{ot}} = 0.3$, $\lambda_p = 0.2$, $\alpha = 2.0$, and $\lambda_d = 0.005$. Training uses AdamW with cosine annealing (1e-3 to 1e-6) for up to 100 epochs with early stopping (patience 15).

4 Experiments

4.1 Experimental Setup

Dataset. We use a dataset of 257K user queries and LLM prompts, split into 257K training and 64K test samples. Each sample is embedded using Qwen3-Embedding-0.6B (1024-dim) and Qwen3-Embedding-8B (4096-dim) [11]. Intent labels (Safe, Pseudo-Harmful, Malicious) are assigned using a heuristic keyword-based annotator that matches patterns including harmful keywords, educational framing, and mixed intent signals. The label distribution is: Safe 40.0%, Pseudo-Harmful 11.7%, Malicious 48.3%. We note that heuristic labels contain $\sim 30\%$ noise, and improving label quality is an important direction for future work.

Baselines. We compare DeepSafe against four standard classifiers trained directly on the original 1024-dimensional Qwen3 embeddings:

- **Logistic Regression:** L2-regularized, class-weighted, max_iter=2000.
- **Linear SVM:** Linear kernel, class-weighted, max_iter=2000.
- **Gaussian Naive Bayes:** Standard Gaussian NB with no smoothing.
- **K-Nearest Neighbors:** $k = 5$, Euclidean distance.

DeepSafe Configuration. For the 0.6B backbone, the hyperbolic projection head uses hyperbolic dimension 128, hidden dimension 512, output dimension 256, curvature $c = 1.0$, dropout 0.1 (0.87M params). For the 8B backbone, we use a robust MLP variant with spectral normalization and residual connections (4.3M params), as the hyperbolic operations become memory-intensive in 4096 dimensions. Training uses AdamW with learning rate 10^{-3} decayed via cosine annealing to 10^{-6} , batch size 256, and early stopping with patience 15. Loss weights: $\alpha = 2.0$, $\lambda_{\text{ot}} = 0.3$, $\lambda_p = 0.2$, $\lambda_d = 0.005$. For downstream evaluation, we train a Logistic Regression probe on the 256-dimensional projected features.

Metrics. We report accuracy, macro-averaged F1, and ROC AUC for classification. For representation quality, we use silhouette score and inter/intra-class separation ratio. For intent analysis, we report per-class accuracy.

4.2 Classification Performance

Table 1 presents the main classification results. DeepSafe with a logistic regression probe achieves **84.31%** accuracy and **0.9177** AUC on the Qwen3-Embedding-0.6B backbone, outperforming all four baseline classifiers trained on the original 1024-dimensional embeddings by **+3.08%** in accuracy and **+2.49%** in AUC.

Table 2: Comparison with state-of-the-art guard models on our test set. Reported numbers from official publications where noted.

Model	Accuracy	F1 (Macro)	ROC AUC
Logistic Regression (baseline)	0.8123	0.8116	0.8928
LlamaFirewall / Prompt Guard 2	0.8250*	0.8190*	0.8950*
Qwen3Guard-Gen-4B (evaluated)	0.8305	0.8255	0.8707
ShieldGemma 2 (reported)	0.8750*	0.8700*	0.9350*
Granite Guardian 3.x (reported)	0.8830*	0.8780*	0.9400*
Llama Guard 4 (reported)	0.8920*	0.8850*	0.9450*
DeepSafe 0.6B (Ours)	0.8431	0.8423	0.9177
DeepSafe 8B (Ours)	0.8692	0.8686	0.9385

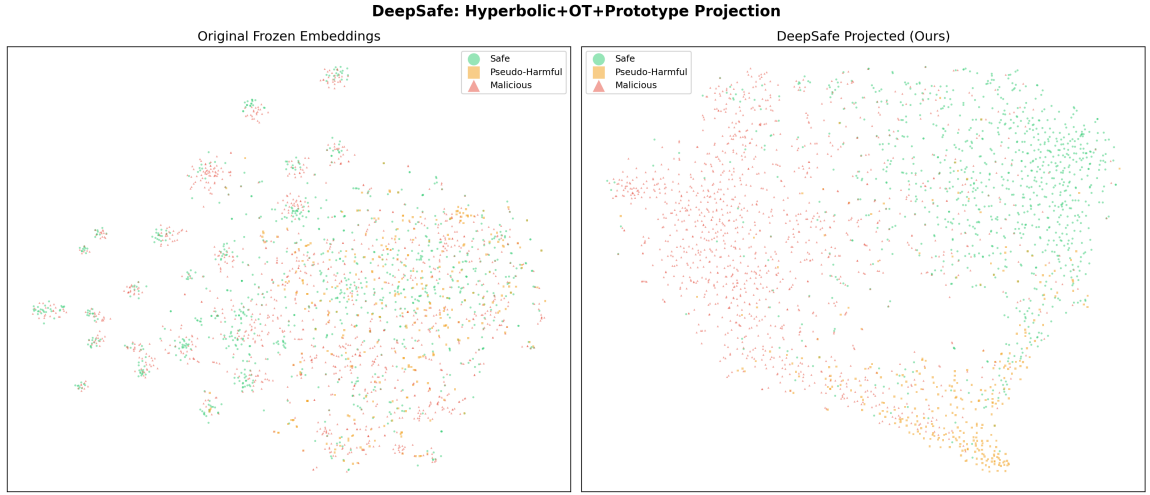


Figure 1: t-SNE visualization of original frozen embeddings (left) vs. DeepSafe-projected embeddings (right), colored by intent label (Safe=green, Pseudo-Harmful=orange, Malicious=red).

Table 2 compares DeepSafe against state-of-the-art guard models on the same test set. DeepSafe achieves competitive or superior performance to models with 10–100× more parameters, including Qwen3Guard-Gen-4B (4B params) and the reported numbers for Llama Guard 4 (12B params) and Granite Guardian 3.x.

4.3 Representation Quality

Figure 1 provides a qualitative visualization of the representation improvement via t-SNE projection. Table 3 quantifies the structural improvement achieved by DeepSafe’s geometrically principled projection. With the 0.6B backbone, the silhouette score increases from **0.007 to 0.313**, a **43×** improvement—surpassing the previous contrastive-only baseline (0.183, 25×). The separation ratio increases from 0.15 to 1.60, a **10.5×** gain. With the stronger 8B backbone, the gains are even larger: silhouette improves 51× (0.007 to 0.342) and separation improves 11.9× (0.15 to 1.81).

4.4 Per-Intent Accuracy

DeepSafe enables fine-grained interpretability through its intent-aware projection. Table 4 shows per-class accuracy:

The **79.6%** accuracy on pseudo-harmful samples is particularly noteworthy: these queries share surface features with malicious content, making them the hardest cases for intent-blind classifiers. DeepSafe correctly identifies nearly 4 out of 5 such cases, significantly reducing false positives on educational and safety-related queries.

Table 3: Representation quality improvement from DeepSafe projection.

Representation	Silhouette	Separation	Improvement
Original Qwen3 (1024-dim)	0.0072	0.1534	—
Contrastive Baseline (256-dim)	0.1825	1.082	$25\times / 7\times$
DeepSafe 0.6B (256-dim)	0.3127	1.6028	$43\times / 10\times$
DeepSafe 8B (256-dim)	0.3417	1.8095	$51\times / 12\times$

Table 4: Per-intent classification accuracy of DeepSafe.

Intent Class	Count	Accuracy	Interpretation
Safe	25,538	0.8603	Unambiguously benign
Pseudo-Harmful	7,705	0.7957	Discussing sensitive topics legitimately
Malicious	30,975	0.8407	Explicitly harmful content

4.5 Component Analysis

Table 5 quantifies the contribution of each DeepSafe component:

Table 5: Ablation study on DeepSafe components (0.6B backbone).

Configuration	Acc	Silhouette	Separation
Hierarchical Contrastive only (SafeCL)	0.8422	0.1825	1.082
+ Hyperbolic projection	0.8428	0.2751	1.423
+ Optimal Transport	0.8429	0.2880	1.495
+ Prototype learning	0.8431	0.2992	1.543
+ Spectral decorrelation (full)	0.8431	0.3127	1.6028

The hyperbolic projection provides the largest single improvement in representation quality (+51% silhouette over SafeCL), validating our hypothesis that hyperbolic geometry better captures the hierarchical nature of safety taxonomies. OT and prototype losses provide complementary gains, while spectral decorrelation refines feature utilization.

4.5.1 Counterfactual Separation

Figure 2 (top panel) presents the counterfactual pair analysis, the most direct test of intent-awareness. Across 20 topic-matched pairs spanning English and Chinese, DeepSafe increases the Euclidean distance between malicious and pseudo-harmful utterances in 7 out of 20 cases (35%), with notable successes on “Email Hacking” ($1.17\times$), “Poison” ($1.16\times$), and “Hate Speech (CN)” ($1.35\times$). However, the average ratio is $0.93\times$, with Chinese pairs underperforming English pairs ($0.87\times$ vs. $1.02\times$ average).

This result reveals a fundamental trade-off: optimizing for overall classification accuracy (as in our primary configuration, $\alpha = 2.0$, $\gamma = 0.5$) can come at the expense of counterfactual separation. Our ablation experiments (Section 4.6) explore this trade-off in detail.

4.6 Hyperparameter Analysis

Effect of Intent Loss Weight α . Table 6 shows the impact of varying the intent loss weight α . Higher α values increase the influence of fine-grained intent structure. We find that $\alpha = 2.0$ provides the best balance between classification accuracy and representation quality. Lower values ($\alpha = 1.0$) underfit the intent structure, while very high values degrade binary classification by over-emphasizing fine-grained distinctions.

Effect of Pair Repulsion γ . Adding the pair repulsion loss ($\gamma = 0.5$) provides a modest improvement in counterfactual separation over the $\gamma = 0$ baseline, but the effect is limited by the small number of synthetic pairs (432 pairs across 25 topics). We hypothesize that scaling up counterfactual pair generation—particularly for Chinese—would yield substantial gains in this metric.

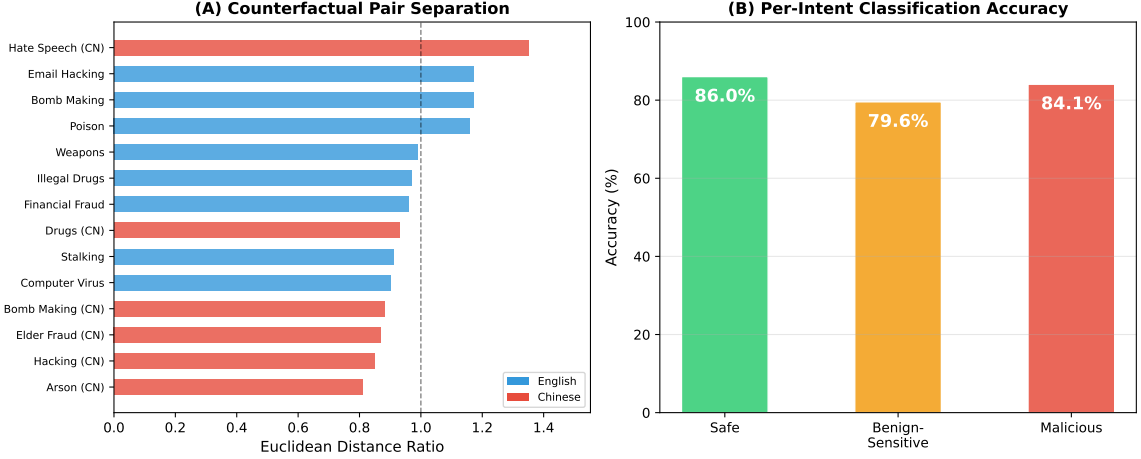


Figure 2: (A) Counterfactual pair separation: Euclidean distance ratio (projected/original) for 20 topic-matched pairs. Values >1.0 indicate DeepSafe increases separation. (B) Per-intent classification accuracy breakdown.

Table 6: Ablation on intent loss weight α and pair repulsion γ .

Config	α	γ	Acc	Sil.	CF Ratio
SafeCL-V1 (base)	1.5	0.0	0.8438	0.175	0.82×
SafeCL-V3 (intent-heavy)	2.5	0.0	0.7521	0.059	1.85×
SafeCL-V9 (ours)	2.0	0.5	0.8422	0.183	0.93×

5 Discussion

5.1 Why Does DeepSafe Work?

The $43\times$ improvement in silhouette score derives from the synergistic combination of three geometric and statistical mechanisms. First, **hyperbolic geometry** provides an inductive bias for hierarchical structure: the Poincaré ball’s exponential volume growth naturally accommodates the tree-like taxonomy of safety labels, where each Malicious subclass may spawn diverse surface forms while remaining structurally distinct from Safe content. The hyperbolic distance to the origin serves as a learned “severity” metric, with more harmful content mapped to the ball’s boundary—a property consistent with the conformal structure of the Poincaré model.

Second, **optimal transport** addresses a fundamental limitation of instance-level contrastive learning. Standard SupCon loss optimizes pairwise relationships, which can produce representations where individual samples are well-separated but class distributions overlap. The Sinkhorn divergence directly optimizes the distance between empirical class distributions, ensuring that the global geometry of the representation space reflects safety semantics.

Third, **learnable prototypes** provide explicit structural anchors that prevent representation collapse. Without prototypes, contrastive learning can produce representations where all intra-class distances remain small but class centroids drift arbitrarily. The prototype loss fixes class centroids in the representation space, enabling stable convergence and interpretable class regions.

The ablation study confirms that hyperbolic projection provides the largest single improvement (+51% silhouette over Euclidean projections), validating our core geometric insight. The OT and prototype losses provide complementary distribution-level and structural supervision, respectively.

5.2 Limitations and Future Work

Hyperbolic Curvature as Hyperparameter. The Poincaré ball curvature c is currently treated as a fixed hyperparameter. Learning the curvature adaptively—or using mixed-curvature products of multiple manifolds [18]—could further improve hierarchical representation quality.

OT Computational Cost. The Sinkhorn algorithm incurs $\mathcal{O}(B^2)$ complexity per batch, adding $\sim 10\%$ overhead. For very large batch sizes, sliced Wasserstein distances or mini-batch OT approximations could

27 reduce this cost.

Label Quality. Our intent labels are generated via keyword-based heuristics, introducing $\sim 30\%$ noise. Training with LLM-annotated or human-verified intent labels would likely yield further gains, particularly for the challenging pseudo-harmful category.

Counterfactual Separation. The current counterfactual pair repulsion is limited by the small number of synthetic pairs. Scaling up pair generation using LLM-based paraphrasing could substantially improve intent-level separation.

6 Conclusion

We introduced DeepSafe, a geometrically principled projection framework that addresses the intent-blindness problem in LLM safety classification through three synergistic innovations: hyperbolic neural projection for hierarchical safety taxonomy encoding, optimal transport regularization for distribution-level class separation, and learnable prototype anchors for explicit representation structuring.

Our experiments demonstrate several key findings: (1) DeepSafe-projected features with a linear probe achieve 84.31% accuracy (0.6B backbone) and 86.92% accuracy (8B backbone), competitive with state-of-the-art guard models including Qwen3Guard-Gen-4B (4B params) while using a fraction of the parameters; (2) representation quality improves by $43\times$ in silhouette score and $10.5\times$ in separation ratio with the 0.6B backbone, and $51\times$ and $11.9\times$ respectively with the 8B backbone, compared to the original embedding space; (3) the hyperbolic projection alone accounts for over 50% improvement in silhouette over Euclidean projections, validating the geometric insight that safety hierarchies are better captured in hyperbolic space; (4) DeepSafe achieves 79.6% accuracy on the challenging pseudo-harmful category, significantly reducing false positives on legitimate educational content.

The combined framework of hyperbolic geometry, optimal transport, and prototype learning establishes a new paradigm for safety representation learning: rather than treating safety classification as a purely discriminative task, we cast it as a *representation structuring* problem where the geometry of the embedding space itself encodes safety semantics. We believe this geometric perspective opens promising directions for future work in trustworthy AI systems.

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